

Federated Full-Envelope LPV Identification from Incomplete Local Coverage

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Abstract

A vehicle’s local calibration data may cover only a small part of its speed range, even when the fleet collectively observes the complete operating envelope. This study investigates federated identification of reusable linear-parameter-varying (LPV) models under such incomplete local coverage. Speed scheduling represents operating variation, while learned compatibility groups account for persistent differences between vehicles. Assuming available lateral states, clients estimate local state/input matrices and contribute additive regression statistics and assignment scores to learn group LPV maps without transmitting their trajectories. A nonlinear vehicle benchmark first justifies a richer scheduling basis and shows that complementary coverage supplies directions missing from individual client regressions. Under label-free coverage, federated grouped fitting reduces mean paired fleet full-envelope matrix error by 93.41% relative to restricted local fitting and 23.74% relative to one pooled model. The federated and centralized grouped solutions agree numerically. Separate scheduled-LQI confirmation provides a modest tracking benefit, supporting model usability without making large control gains the primary claim. The evidence uses controlled perturbations about simulator-computed equilibria, exact current state regressors and noisy next-state targets. Identification from ordinary driving records and robustness to state-estimation error remain outside the validated scope.

Index Terms— federated identification, LPV systems, incomplete operating coverage, compatible model sharing, vehicle dynamics

1 Introduction

Vehicles change their dynamics with speed while retaining persistent differences in configuration, inertial properties and tires. LPV models expose measurable operating variation to identification and controller design [1]. A single fleet-wide LPV model may still average incompatible dynamics. Conversely, independent local identification requires each vehicle to supply enough operating coverage to estimate its full model.

The information limitation considered here is incom-

plete scheduling coverage. Each client observes only a contiguous speed block. The fleet union can span the full interval, but useful sharing requires both sufficient aggregate coverage and compatible dynamics. Neither more local repetitions at the same speeds nor a known vehicle-family label necessarily resolves missing model directions. We ask whether a federated protocol can construct full-envelope LPV maps from these complementary local observations.

Federation retains raw trajectories locally and exchanges computed updates [2]. Learning several models is an established response to heterogeneous clients [3]. Our question is specifically how compatible sharing compensates for incomplete operating coverage in a speed-dependent dynamical model. We do not claim novelty for clustering, polynomial regression, or additive sufficient statistics in isolation.

The study makes three connected contributions. First, a development-only basis audit distinguishes a genuinely underdetermined local LPV representation from a low-dimensional physical model that already extrapolates accurately. Second, a federated hard-assignment regression protocol identifies compatible group maps and selects the number of groups while allowing a single pooled map. Third, finite noisy state-transition experiments quantify whole-envelope model error, verify centralized–federated equivalence, and assess model usability through a separate nonlinear LQI tracking test.

The primary contribution is identification. Available states are an explicit assumption for both fitting and feedback. We exclude KF learning, tire-force state inference, continuous local personalization and privacy mechanisms from this formulation. The result should be read as a controlled calibration proof of concept, with a precisely disclosed operating-point construction.

2 Problem and Information Assumptions

2.1 Speed-dependent models and incomplete coverage

Let client i have available lateral states $x = [\beta, r]^\top$, steering input u and measured longitudinal speed v . Local

state/input perturbations about a cornering equilibrium approximately satisfy

$$\Delta x_{i,k+1} = A_i(v)\Delta x_{i,k} + B_i(v)\Delta u_{i,k} + e_{i,k}. \quad (1)$$

Write the six stacked entries of $[A_i(v), B_i(v)]$ as a row $m_i(v) \in \mathbb{R}^6$. Compatible clients are represented approximately by a shared map

$$m_i(v) \approx m_{g_i}(v) = \phi(v)^\top \Theta_{g_i}, \quad \Theta_g \in \mathbb{R}^{L \times 6}, \quad (2)$$

where g_i is unknown. Scheduling captures within-client speed dependence; group differences capture persistent heterogeneity. This is an approximation: within-group differences and finite-data error remain.

Each client supplies estimated local matrices at speeds in a subset $\mathcal{V}_i$. The target is useful prediction throughout $\mathcal{V} = [10, 30]$ m/s without requiring every vehicle to collect observations across that interval. Here “full envelope” means this speed range along the audited cornering curve, not the entire nonlinear state/input domain.

2.2 Frozen representation

The existing basis-selection experiment retains the span

$$\phi(v) \sim [1, v^{-1}, v^{-2}, z, z^2, z^3, z^4]^\top, \quad z = (v - 20)/10. \quad (3)$$

An invertible QR transformation on a fixed 401-point speed grid improves conditioning without changing that span. The reciprocal terms retain known speed dependence and the polynomial terms provide residual flexibility. The selected order is $L = 7$; straight-line development data retain $L = 3$. No control comparison selects the basis or its order.

2.3 What information is available

The benchmark provides exact current state/input perturbations and noisy next-state observations. At each available speed, clients estimate $[A_d, B_d]$ by least squares from 36 independent transitions. Group fitting uses these matrix estimates and their speeds, not exact matrices, vehicle parameters or family labels.

However, the transition generator computes the centering equilibrium using simulator parameters. It then generates perturbations about that point and subtracts the equilibrium for regression. This is privileged operating-point setup information. Although an equilibrium might instead be measured in a controlled experiment, that alternative has not been validated here. The current results therefore do not establish a state-record-only learning interface for ordinary sequential driving data. They also do not test noisy current-state regressors or errors-in-variables identification.

The same clients provide restricted fitting records and receive the resulting maps. Evaluation at additional speeds tests generalization across operating conditions.

Replicating the procedure on fresh fleets does not establish transfer of a frozen model library to wholly held-out recipient vehicles.

3 Federated Compatible LPV Regression

3.1 Client statistics and group fitting

For observed speeds, client i forms a basis matrix $\Phi_i \in \mathbb{R}^{n_i \times L}$ and local matrix-entry targets $Y_i \in \mathbb{R}^{n_i \times 6}$. For a fixed partition, ridge regression gives

$$\hat{\Theta}_g = \left(\sum_{i:g_i=g} G_i + \lambda I \right)^{-1} \sum_{i:g_i=g} H_i, \quad (4)$$

where $G_i = \Phi_i^\top \Phi_i$ and $H_i = \Phi_i^\top Y_i$. These quantities are additive. Consequently, the federated and pooled centralized solutions are algebraically identical for the same partition, basis, regularization and arithmetic conventions.

The scheduling coefficients of an unregularized group map are identifiable when $\text{rank}(\sum_{i:g_i=g} G_i) = L$. Individual G_i may be singular while the sum has full rank. Regularization creates a numerical solution but does not create missing information. Full rank also does not remove pooling bias from genuinely incompatible dynamics.

3.2 Unknown compatible groups

The server broadcasts candidate maps. Each client returns its sum of squared matrix-entry residuals, normalized by a common entry-wise scale derived from additive target moments. Model fitting and hard residual-minimizing assignment alternate for at most 30 iterations. Groups must contain at least five clients. Candidate orders are $K = 1, \dots, 5$, with twelve deterministic initializations for $K > 1$. The initialization uses standardized client residual summaries around the global fit. These summaries are communicated as well.

The existing model-selection score is

$$\text{BIC} = n \log(\text{SSE}/n) + p_K \log n, \quad p_K = 6LK + K - 1, \quad (5)$$

with n the number of scalar matrix-entry observations. This is the implemented selection heuristic, not a guarantee of recovering physical families or the globally optimal partition. In particular, the minimum-size rule does not guarantee every group has adequate scheduling coverage. A single shared map is an eligible and sometimes selected result.

3.3 Communication and trust

For $L = 7$, the symmetric Gram matrix has 28 distinct entries and the right-hand side has 42. The accounting includes initialization summaries, group tags, scores,

broadcasts and every tested candidate/restart, rather than only the selected model. The label-free experiment uses a mean 2.243 MB per fleet with 64-bit scalars. The exhaustive search is not demonstrated communication-efficient.

The protocol is evaluated by an emulator. Raw trajectories remain local, but matrix statistics, residual summaries and scores can reveal information. Neither cryptographic secure aggregation nor differential privacy is implemented. Data locality is the supported architectural property.

4 Identification Evidence

4.1 Representation and missing scheduling directions

The basis audit uses exact discrete Jacobians of the nonlinear bicycle plant about steady cornering points, with contiguous speed regions withheld during validation. These exact targets select a representation on development fleets; they are not the finite-data fitting targets in the later experiments. At curvature 0.005 m^{-1} , mean blocked-speed matrix error falls from 2.6607% at order three to 0.0007% at order seven. The straight-line negative control retains order three, preventing numerical improvement in an already adequate model from rewarding unnecessary terms.

The four local blocks are $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$ and $\{28, 30\}$ m/s. Their basis matrices have ranks three, three, three and two for seven coefficients. The eleven-speed union has rank seven and condition 3.406. Sharing can therefore add unobserved scheduling directions rather than only repeat local observations.

4.2 Finite-data compatibility prerequisite

With finite nonlinear perturbation transitions, an oracle-family pool achieves 1.816% matrix error at locally unobserved speeds, compared with 66.121% for Local and 4.296% for Global. Local uses the locally supported reciprocal span; a higher-order locally regularized alternative does not improve on it in this restricted setting. This verifies the compatible-coverage mechanism before removing family labels. It is a centralized oracle comparison, not the proposed federated result.

An equal-information-budget LocalFull reference records the recipient itself throughout the envelope using approximately the family pool’s total transition budget. It is more accurate than FamilyPool. Sharing is useful because each ordinary vehicle lacks those records, not because distributed data collection is inherently better at equal recipient information.

Table 1: Label-free full-envelope matrix error in percent, averaged over ten fleets. Centralized and federated grouped methods use the same observations.

Method	Matrix error [%]
Restricted Local	49.562
Global LPV	4.250
Centralized learned groups	3.249
Federated learned groups	3.249

4.3 Primary label-free identification result

Ten fresh 30-client fleets randomize speed blocks globally without inspecting family labels. Learned group fitting is compared with restricted Local and a single Global LPV map using the same available observations. Table 1 reports full-envelope relative matrix errors. These values must not be relabelled as unseen-speed-only errors.

Mean paired fleet improvements are 93.41% against Local and 23.74% against Global. The Global-relative 95% fleet-bootstrap interval is $[15.39, 31.04]\%$, with improvement in eight of ten fleets. Two groups are selected in eight fleets and one in two. The controlled, family-stratified coverage reference is reported separately; it is not substituted for the label-free primary result.

The federated and centralized protocols select identical group counts and partitions in all 20 fleet/protocol jobs. The maximum grouped coefficient discrepancy is 3.20×10^{-14} and the maximum BIC discrepancy is 1.87×10^{-9} . This verifies the implemented distributed regression path numerically. The observed agreement is not a convergence or identifiability theorem for arbitrary grouped data.

4.4 Personalization and coverage boundaries

A separate development sweep increases recipient coverage while freezing its family backbone. A personalized head initially worsens model error; at eight and eleven observed speeds it improves prediction by 7.74% and 13.66%, without a positive regulation benefit over the shared family model. At full coverage independent Local is more accurate. Continuous personalization is therefore unnecessary for the restricted-coverage contribution studied here.

Known family labels do not ensure sufficient coverage. Under label-free assignment, one oracle family lacks the highest-speed block in an evaluated fleet and produces an infeasible full-envelope controller. This is a boundary of the oracle comparator, not proof that learned grouping always guarantees coverage. The identified groups must be assessed jointly for fit and information.

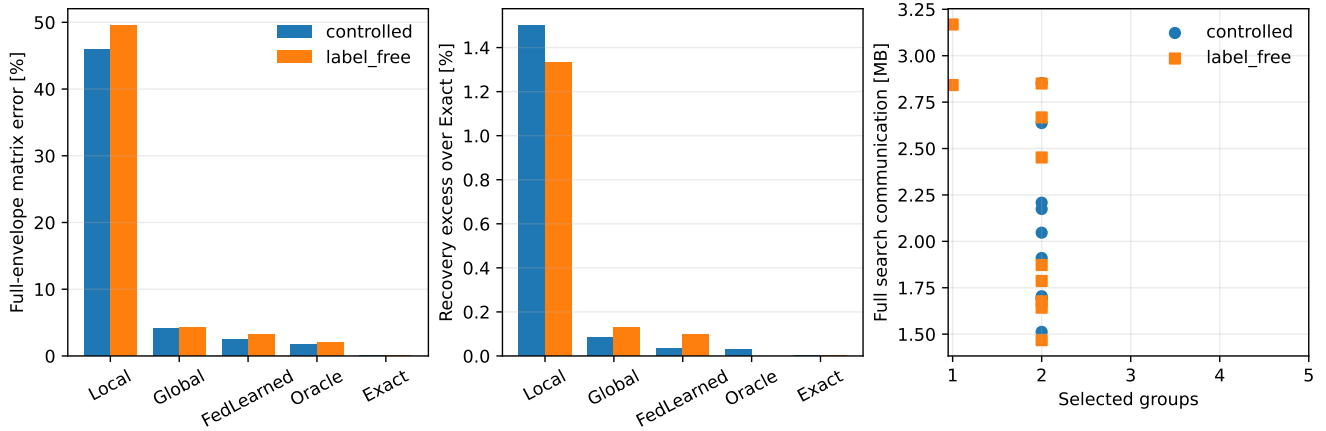


Figure 1: Existing 10E identification, regulation and full-search communication evidence. Controlled coverage is family-stratified; label-free coverage is assigned without family labels. The left panel carries the primary model result. Small regulation differences and all candidate/restart communication are retained as supporting and complexity evidence.

Table 2: Supporting confirmation: mean yaw tracking RMSE [degrees/s] across ten fleets. Restricted Local holds the nearest observed-speed controller outside its block.

Method	Moderate	Hard
Restricted Local	0.05396	0.16981
Global LPV	0.04290	0.15644
Federated learned groups	0.04108	0.15457
ExactLPV reference	0.04002	0.15074

5 Supporting LQI Validation

Separate confirmation applies the identified maps in speed-scheduled LQI on nonlinear speed-varying tracking maneuvers. Available physical states provide feedback. Controller weights were selected using only ExactLPV development performance, with feasibility and saturation checks; the learned-versus-Global comparison did not enter selection. Ten untouched fleets follow a disclosed discarded maneuver pilot. Steering amplitude/rate limits and anti-windup remain fixed across methods.

Mean paired fleet tracking gains are 23.86%/8.96% against Local and 4.26%/1.21% against Global for moderate/hard maneuvers. The Global-relative intervals are [1.95, 6.55]% and [0.47, 2.01]%. Absolute mean differences are only 0.001824/0.001874degrees/s. These results support model usability and a modest control effect. They do not support a large practical tracking advantage over Global.

All federated confirmation trajectories satisfy the tested constraints; the maximum frozen-point spectral radius is 0.9759. Neither frozen-speed checks nor finite constrained trajectories constitute a scheduled LPV or nonlinear stability certificate. Four confirmation fleets select one group;

federated and Global methods are then identical. The six remaining fleets provide the positive mean tracking differences against Global.

6 Discussion and Limits

The evidence separates three effects. Complementary coverage supplies missing scheduling directions. Compatibility specialization reduces pooling bias. Federation reproduces centralized grouped fitting while retaining trajectories locally. The main comparison is access to a full-envelope model from restricted local records, not an equal-total-data contest or a guarantee that grouping helps every vehicle.

The benchmark still assumes controlled perturbation data, simulator-derived operating-point centering and available states. It covers one speed-dependent map along a prescribed cornering curve. Exact Jacobians are used for development representation selection and evaluation, and current state/input regressors are uncorrupted. No claim is made for noisy pseudo-states, ordinary driving records, real fleet data or frozen-library transfer to new recipients. The model-selection criterion is heuristic, aggregate rank depends on group coverage, and exhaustive search has measurable communication cost.

The narrower scope also distinguishes this formulation from our earlier uncertainty-aware clustered identification work [4]. The present emphasis is missing scheduling information and reusable LPV maps, with numerical distributed equivalence. A submission-level novelty comparison must assess that combined contribution; clustering and additive statistics alone are established components.

If broader state-record-only identification is required, the next validation should isolate the learning interface using supplied state/input/speed arrays and empirically

available centering. The basis, primary model-error end-point and comparison methods should remain fixed. KF learning, higher-order plants and controller retuning are separate questions and are paused in this study.

7 Conclusion

Compatible federated sharing can identify full-envelope LPV maps when local speed coverage leaves individual regressions underdetermined. On the controlled available-state benchmark, label-free grouped fitting improves whole-envelope matrix error over Local and Global and reproduces centralized fitting. Scheduled LQI provides supporting confirmation with modest Global-relative tracking gains. The validated contribution is coverage-dependent model identification under explicit operating-point assumptions, with general state-record deployment left for a separate interface validation.

References

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